lime or lime with surkhi in different proportions. Lime mortar has the properties like high plasticity, can be placed easily, good cohesiveness, durability and hardens slowly. It is generally used in lightly loaded parts of building above ground level.

4.2.4.2 Cement Mortar

In cement mortar, cement is used as binding material. Proportion of cement to sand varies from 1:2 to 1:6 or more. Cement mortar is used where a mortar of high strength and water resisting property are required such as underground construction, water saturated soil, etc.

4.2.4.3 Combination Mortar or Gauged Mortar

To improve the quality of lime mortar, cement is sometimes added to it. This is known as gauging. It makes lime mortar more strong and dense.

4.2.4.4 Mud Mortar

In this type of mortar clay and sand are mixed together and used in the construction of temporary sheds.



- Gypsum based mortars have already been used 10,000 years ago.
- Lime came into operation in 6000 BC but it had been used first by romans as a constituent to produce mortar.

4.2.5 Preparation of Cement Mortar

Cement Mortar needed for small works is mixed by hand and for large works is mixed by mixer machine.

4.2.5.1 Hand Mixing



Cement and Sand



Dry Mixing



Wet Mixing



Prepared Mortar

Sand is measured by boxes and cement by weight of bags. The specified quantity of sand is spread first and then cement is spread over it. They are dry mixed again and again

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